WTW 158 Trig worksheet

A. Complete:

1. $\cos\frac{\pi}{4} =$
2. $\sin\frac{3\pi}{4} =$
3. $\cos\frac{7\pi}{4} =$
4. $\sin(-\frac{\pi}{3}) =$
5. $\cos\frac{5\pi}{6} =$
6. $\tan\frac{2\pi}{3} =$
7. $\sin(-\frac{5\pi}{2}) =$
8. $\sin\frac{11\pi}{4} =$
9. $\cos(-\frac{4\pi}{3}) =$
10. $\cot\frac{\pi}{4} =$
11. $\cos\frac{13\pi}{4} =$
12. $\sec\frac{\pi}{4} =$

B. Find $\theta \in [0, 2\pi)$ such that:

1. $\sin\theta = \frac{1}{\sqrt{2}}$
2. $\cos\theta = -\frac{1}{\sqrt{2}}$
3. $\sin\theta = \frac{1}{2}$
4. $\cos\theta = \frac{\sqrt{3}}{2}$
5. $\cos\theta = -\frac{\sqrt{3}}{2}$
6. $\sin\theta = -1$
7. $\sin\theta = -\frac{1}{\sqrt{2}}$
8. $\cos\theta = \frac{1}{2}$
9. $\cos\theta = 0$
10. $\tan\theta = -1$
11. $\sec\theta = -2$
12. $\sin\theta = -\frac{1}{2}$

A.

1. $\frac{1}{\sqrt{2}}$ 2. $\frac{1}{\sqrt{2}}$ 3. $\frac{1}{\sqrt{2}}$ 4. $-\frac{\sqrt{3}}{2}$ 5. $-\frac{\sqrt{3}}{2}$ 6. $-\sqrt{3}$

7. -1 8. $\frac{1}{\sqrt{2}}$ 9. $-\frac{1}{2}$ 10. 1 11. $-\frac{1}{\sqrt{2}}$ 12. $\sqrt{2}$

B.

1. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$ 2. $\theta = \frac{3\pi}{4}$ or $\theta = \frac{5\pi}{4}$ 3, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$ 4. $\theta = \frac{\pi}{6}$ or $\theta = \frac{11\pi}{6}$

5. $\theta = \frac{5\pi}{6}$ or $\theta = \frac{7\pi}{6}$ 6. $\theta = \frac{3\pi}{2}$ 7. $\theta = \frac{5\pi}{4}$ or $\theta = \frac{7\pi}{4}$ 8. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

9. $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$ 10. $\theta = \frac{3\pi}{4}$ or $\theta = \frac{7\pi}{4}$ 11. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$ 12. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

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